Evidencias - Tarea 10

Nombre: Walter Isai Villagran Garcia

Carné: 9960-21-8603

Curso: Programación II

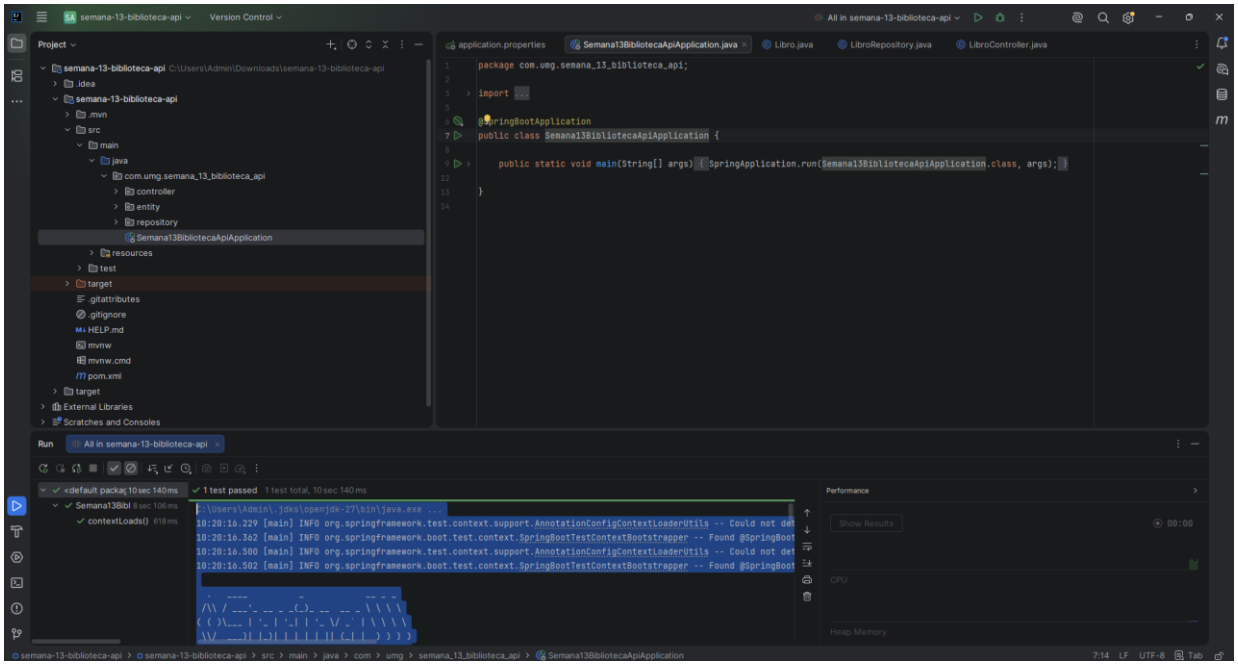
Fecha: 16 de octubre de 2026

Universidad Mariano Gálvez de Guatemala

Enlace del repositorio
https://github.com/Walter1612/Tarea-10-Arquitectura-por-capas

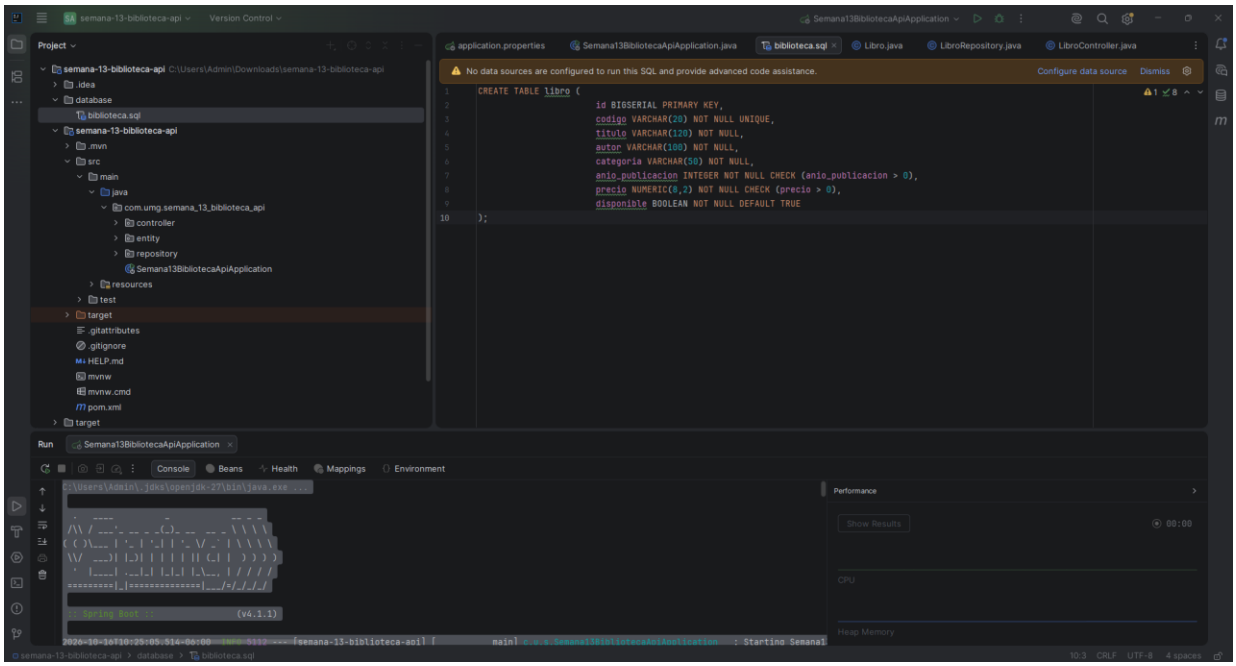
Objetivo
Desarrollar una API REST para gestionar libros mediante una arquitectura por capas, utilizando Spring Boot, Spring Data JPA y PostgreSQL. Se implementaron operaciones CRUD y consultas adicionales, verificadas mediante Postman.

Estructura del proyecto
El proyecto conserva una separación por capas mediante los paquetes controller, entity y repository, además del archivo principal de Spring Boot y la configuración de recursos.



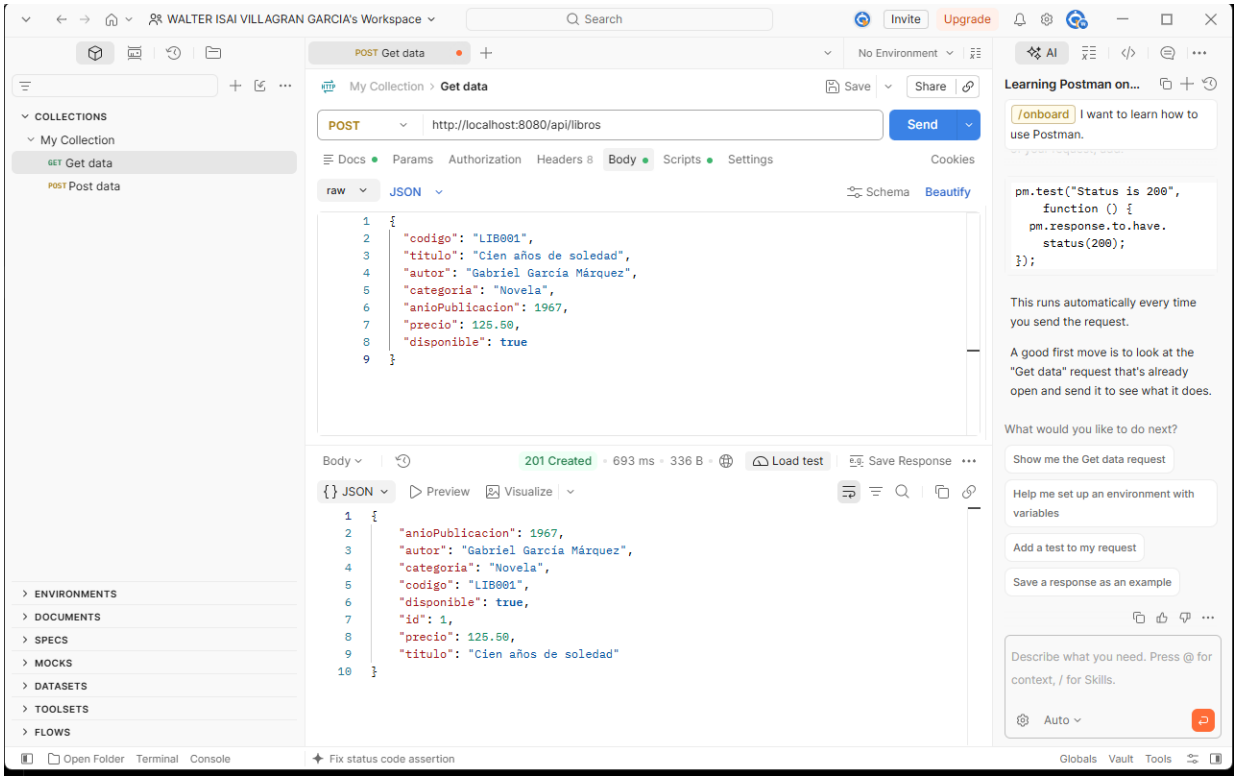
Script de base de datos

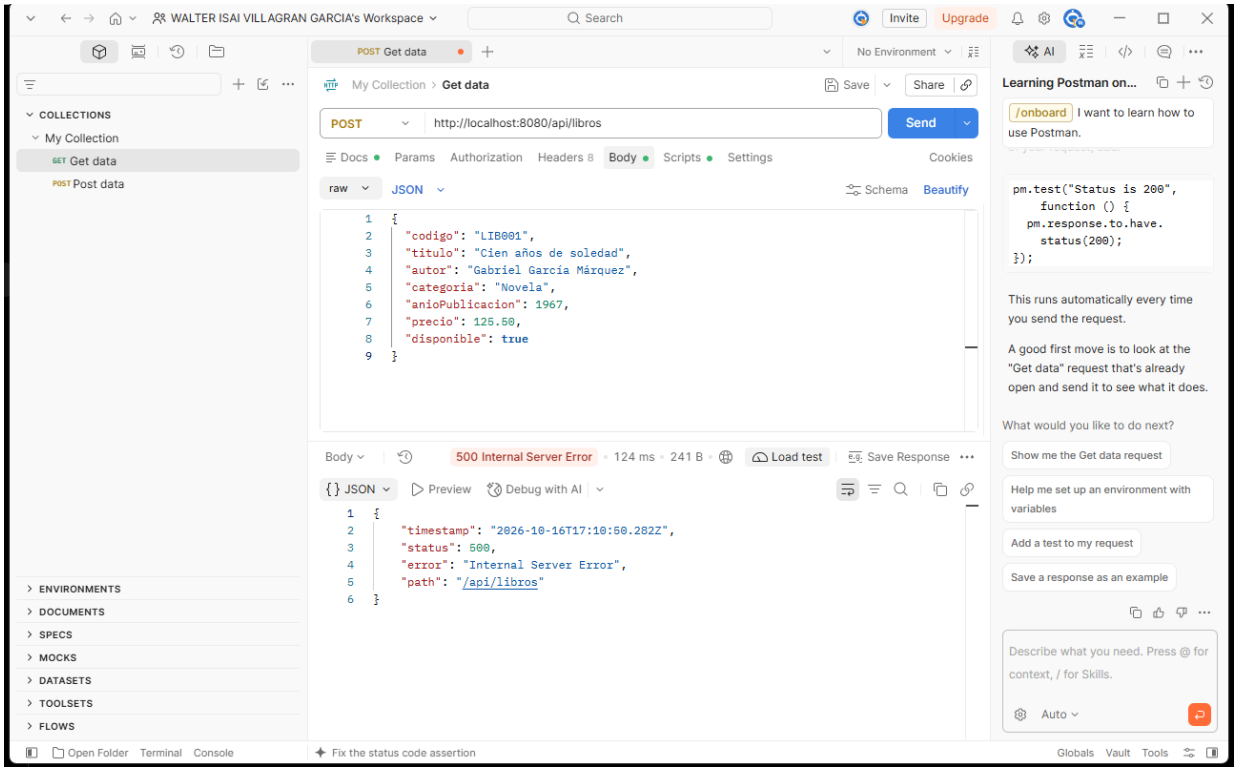
La tabla libro fue definida con identificador autoincremental, código único, datos bibliográficos, precio y disponibilidad.

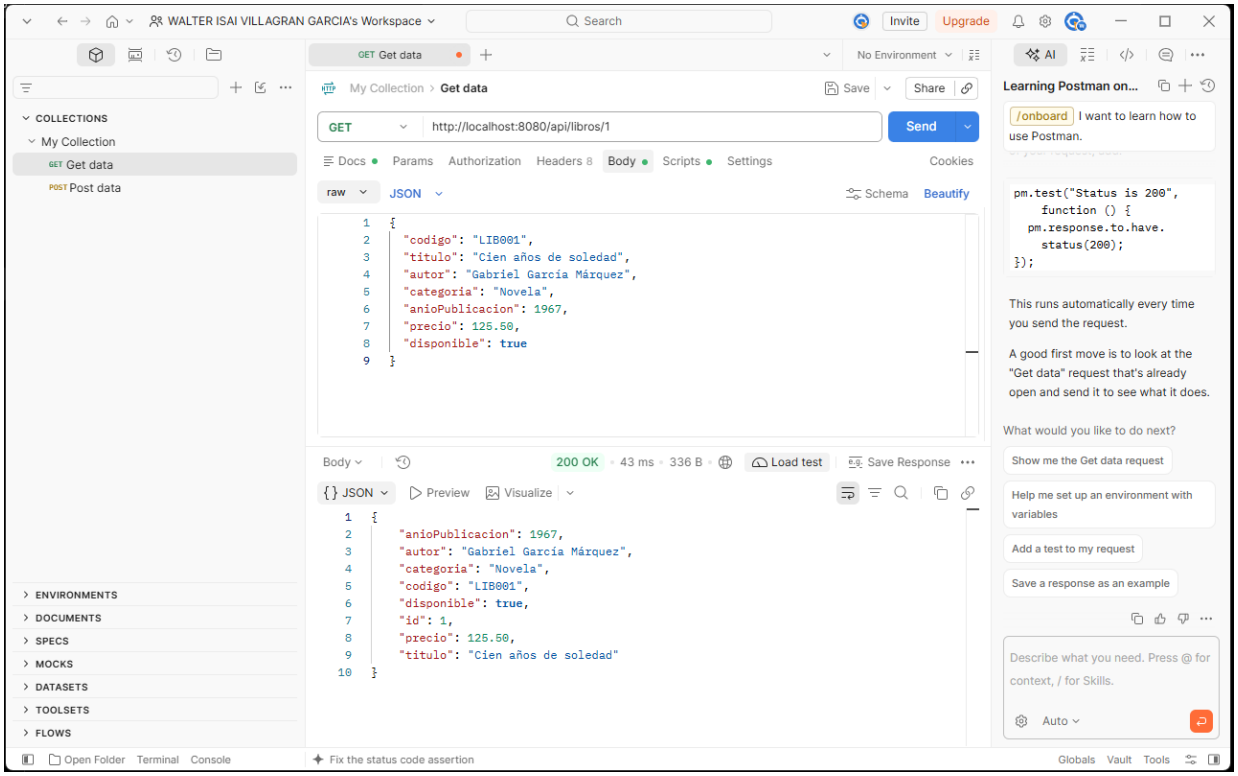


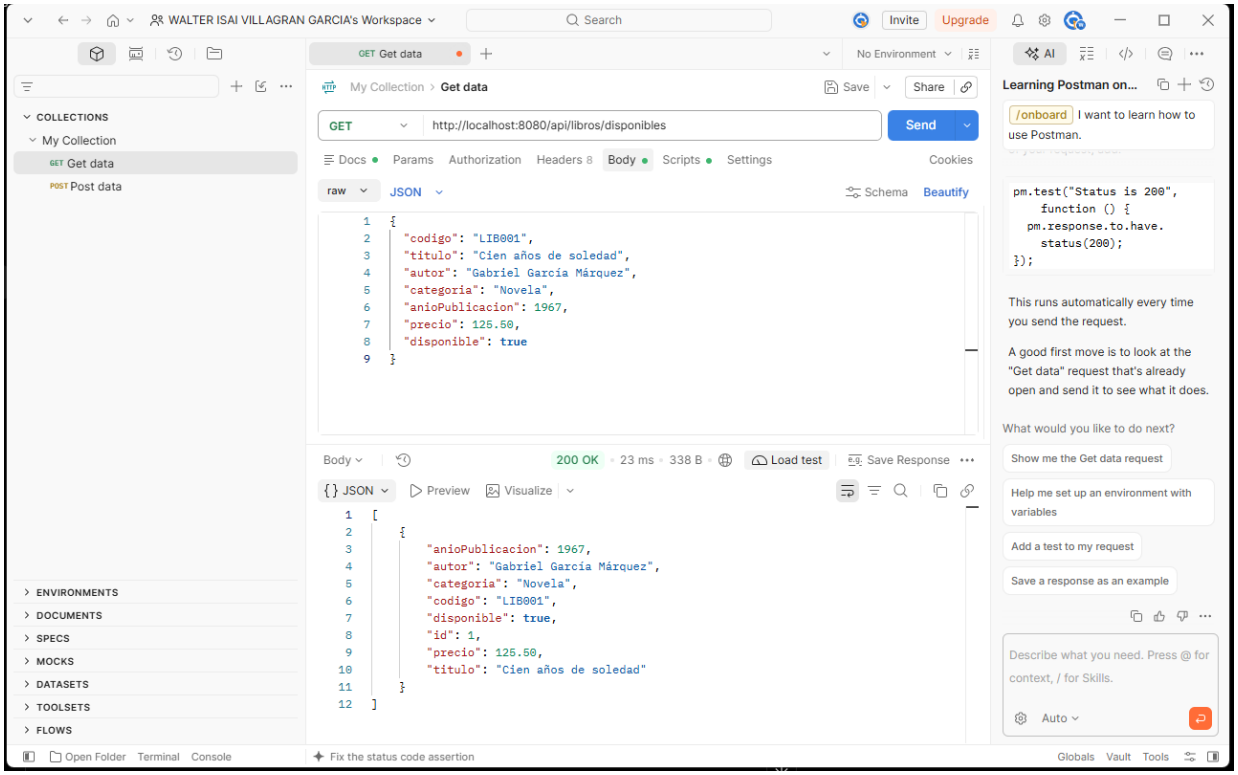
Pruebas de la API con Postman

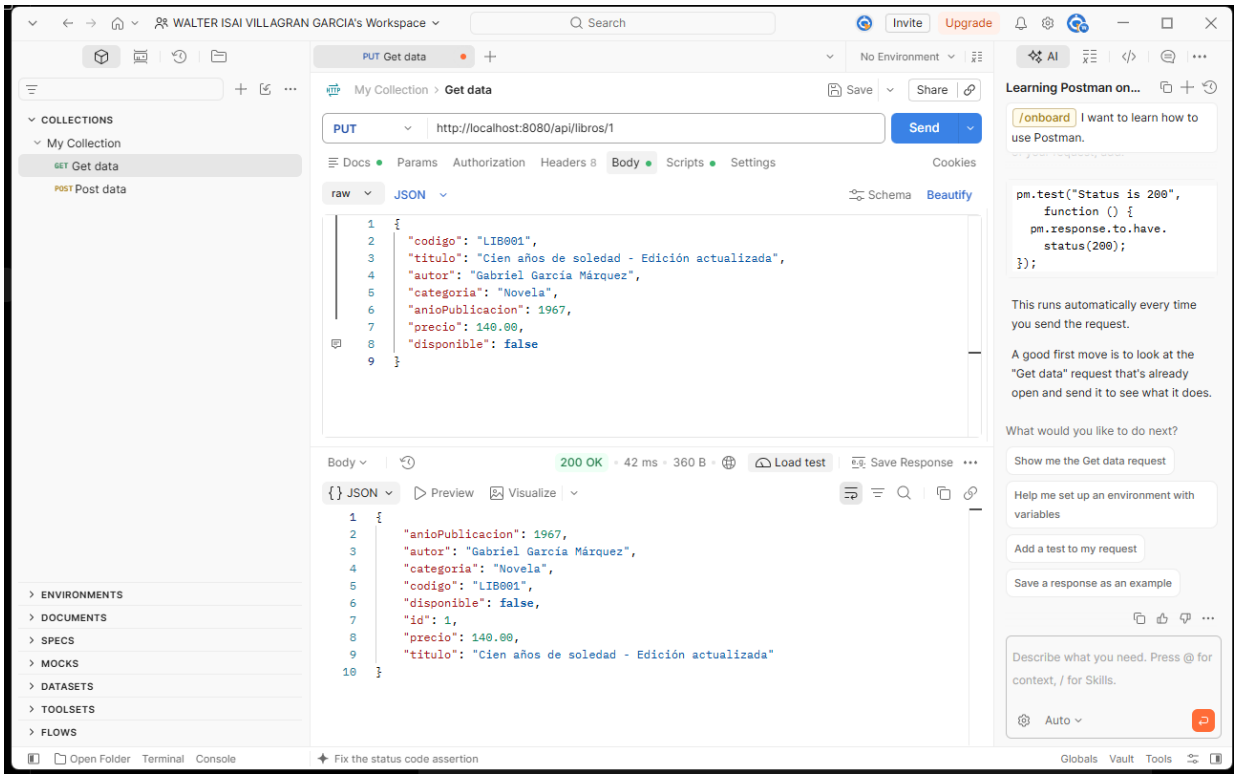
Se realizaron pruebas directas contra los endpoints de la API para comprobar el registro, consulta, actualización, filtros y eliminación de libros.

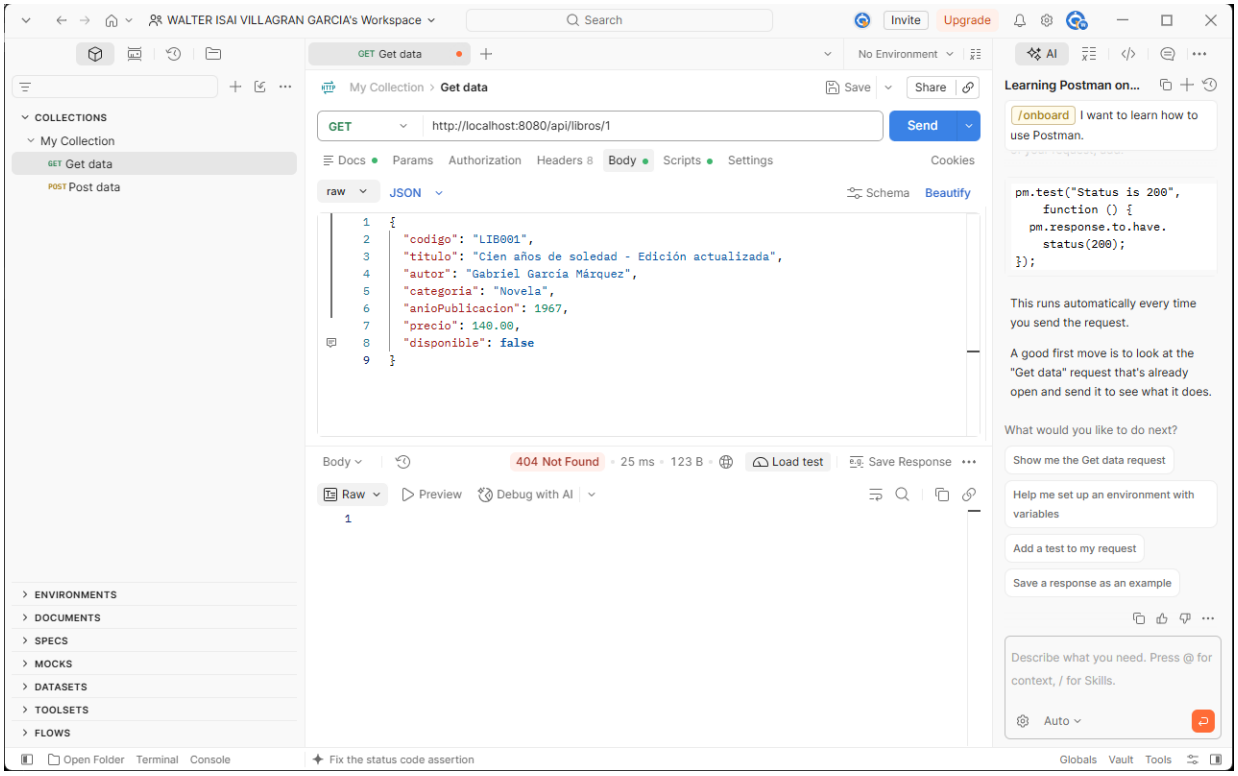



WALTER ISAI VILLAGRAN GARCIA's Workspace
Search
Invite
Upgrade
POST Get data
No Environment
My Collection > Get data
Save
Share
POST
http://localhost:8080/api/libros
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"codigo": "LIB001",
"titulo": "Cien años de soledad",
"autor": "Gabriel García Márquez",
"categoria": "Novela",
"anioPublicacion": 1967,
"precio": 125.50,
"disponible": true
}
Body
500 Internal Server Error
124 ms
241 B
Load test
Save Response
JSON
Preview
Debug with AI
{
"timestamp": "2026-10-16T17:10:50.282Z",
"status": 500,
"error": "Internal Server Error",
"path": "/api/libros"
}
COLLECTIONS
My Collection
GET Get data
POST Post data
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Open Folder
Terminal
Console
Fix the status code assertion
AI
Learning Postman on...
/onboard I want to learn how to use Postman.
pm.test("Status is 200", function () {
pm.response.to.have.status(200);
});
This runs automatically every time you send the request.
A good first move is to look at the "Get data" request that's already open and send it to see what it does.
What would you like to do next?
Show me the Get data request
Help me set up an environment with variables
Add a test to my request
Save a response as an example
Describe what you need. Press @ for context, / for Skills.
Auto
Globals
Vault
Tools


WALTER ISAI VILLAGRAN GARCIA's Workspace
Search
Invite
Upgrade
GET Get data
No Environment
My Collection > Get data
Save
Share
GET
http://localhost:8080/api/libros/1
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
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"autor": "Gabriel García Márquez",
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Body
200 OK
43 ms
336 B
Load test
Save Response
JSON
Preview
Visualize
{
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"codigo": "LIB001",
"disponible": true,
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Upgrade
GET Get data
No Environment
My Collection > Get data
Save
Share
COLLECTIONS
My Collection
GET Get data
POST Post data
GET
http://localhost:8080/api/libros/disponibles
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
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}
Body
200 OK
23 ms
338 B
Load test
Save Response
JSON
Preview
Visualize
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Send
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Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
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"autor": "Gabriel García Márquez",
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"anioPublicacion": 1967,
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"disponible": false
}
Body
200 OK
42 ms
360 B
Load test
Save Response
JSON
Preview
Visualize
{
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DOCUMENTS
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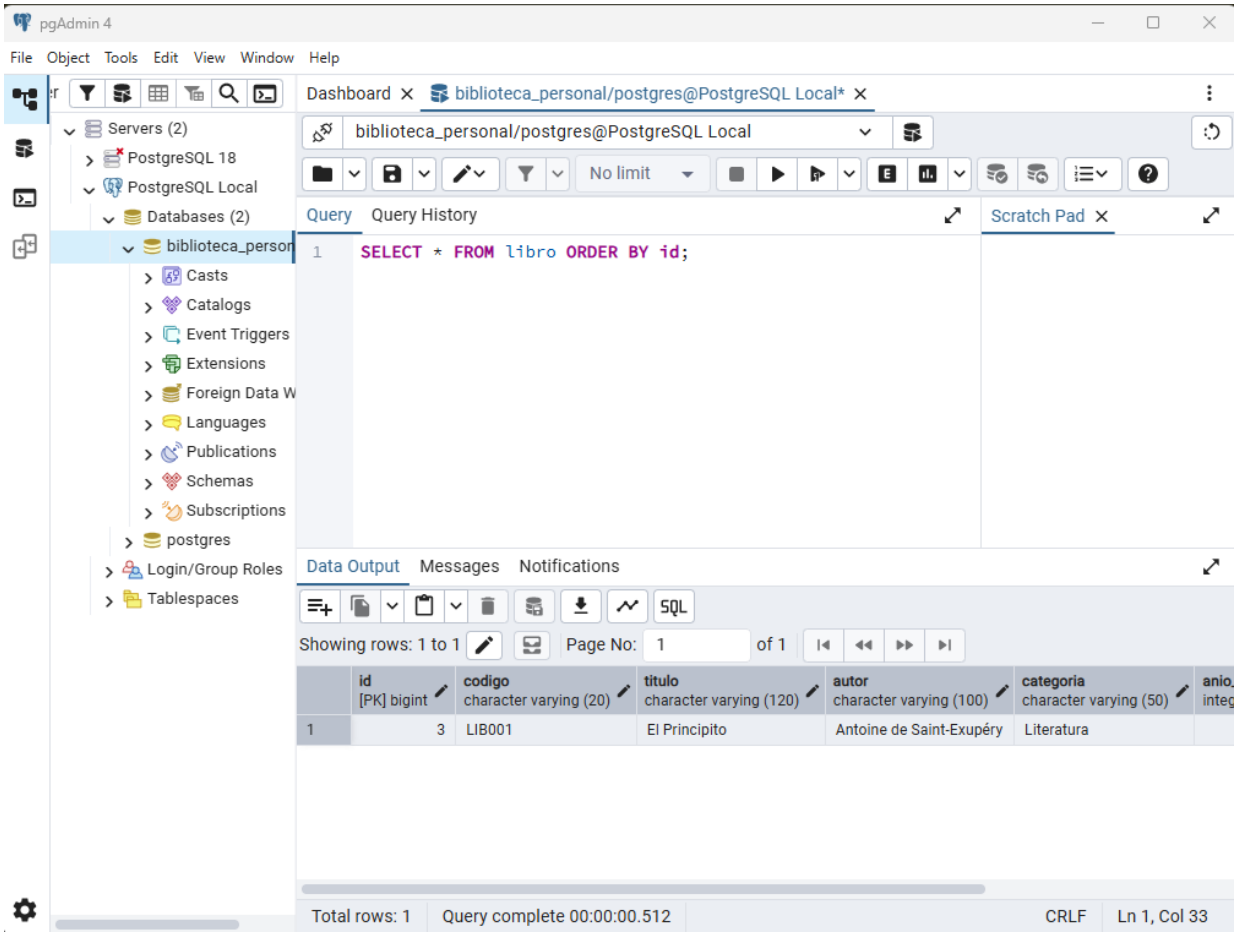


Resultados obtenidos

Elemento	Resultado
POST /api/libros	201 Created - libro registrado correctamente.
GET /api/libros	200 OK - listado recuperado.
GET /api/libros/{id}	200 OK cuando existe y 404 después de eliminar.
GET por categoría	200 OK - filtro por categoría funcionando.
GET /disponibles	200 OK - solo libros disponibles.
PUT /api/libros/{id}	200 OK - datos actualizados.
DELETE /api/libros/{id}	204 No Content - eliminación correcta.

Evidencia en base de datos

La aplicación se conectó correctamente a PostgreSQL y utilizó la base biblioteca_personal. Los cambios realizados desde la API se reflejaron en la tabla libro.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard
biblioteca_personal/postgres@PostgreSQL Local*
Servers (2)
PostgreSQL 18
PostgreSQL Local
Databases (2)
biblioteca_person
Casts
Catalogs
Event Triggers
Extensions
Foreign Data W
Languages
Publications
Schemas
Subscriptions
postgres
Login/Group Roles
Tablespaces
biblioteca_personal/postgres@PostgreSQL Local
No limit
Query
Query History
Scratch Pad
1 SELECT * FROM libro ORDER BY id;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1 of 1
id [PK] bigint
codigo character varying (20)
titulo character varying (120)
autor character varying (100)
categoria character varying (50)
1 3 LIB001 El Principito Antoine de Saint-Exupéry Literatura
Total rows: 1
Query complete 00:00:00.512
CRLF
Ln 1, Col 33